Fiscal Year:	FY 2024	**Task Last Updated:**	FY 11/11/2023
PI Name:	Lynch, Kennda Ph.D.		
Project Title:	Plant Trek: Investigating Strategies for Regolith Pre-Conditioning to Support the Establishment of Plant-Microbe Systems in Martian Habitats		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	77058-1113	**Congressional District:**	36
Comments:	Former email: klynch@lpi.usra.edu		
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBPS E.9: Plant Studies
Start Date:	01/10/2023	**End Date:**	01/09/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:	Ruby, Anna Maria	**Contact Phone:**	321-867-7065
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 01/09/2025 per E. Blancaflor/NASA and NSSC information (Ed., 9/29/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Loureiro, Rafael Ph.D. (Winston-Salem State University (Inc)) Simpson, Anna Ph.D. (Jet Propulsion Laboratory) Venkateswaran, Kasthuri Ph.D. (Jet Propulsion Laboratory)		
Grant/Contract No.:	80NSSC23K0400		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A critical component to long-duration deep space exploration, specifically on Mars, is developing self-sustainable in situ food production and life support systems. To accomplish this task, it will be necessary to understand how to integrate plant-microbe systems optimally with planetary in situ resources. The overarching goal of this proposed Early Career pilot study is to develop and assess an integrated system approach for pre-conditioning and structing Martian regolith into agriculturally stable and usable soil to support plant growth, sustain microbe-plant interactions, minimize plant stress, and optimize food production and life support. As a part of this study, we will test a microbial consortium derived from a natural perchlorate-reducing system as a pre-inoculant for mitigating perchlorate toxins in Martian regolith simulant. We will also evaluate the phased approach of introducing pioneer species and plant-beneficial micro-organisms as an intermediate structure building step for transforming Martian regolith into a viable agricultural substrate that will reduce plant stress and increase seed germination rates and overall biomass production. The results of this study will pave the way for long-term sustainable crop production in a Martian habitat.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As we continue to realize the effects of climate change, the implications for food security long term are starting to become clear. The work done in Plant Trek will help benefit life on Earth and it will help us to understand the fundamental tools for "manufacturing" healthy stand alone agriculture systems should it be necessary to implement closed loop agriculture in areas of planet Earth where the natural environment have become unsuitable for traditional agriculture.
Task Progress:	A critical component to long-duration deep space exploration, specifically on Mars, is developing self-sustainable in situ food production and life support systems. To accomplish this task, it will be necessary to understand how to optimally integrate plant-microbe systems with planetary in situ resources. The overarching goal of this pilot study is to develop and assess an integrated system approach for pre-conditioning and structuring Martian regolith into agriculturally stable and usable soil to support plant growth, sustain microbe-plant interactions, minimize plant stress, and optimize food production and life support. As a part of this study, we will test a microbial consortium derived from a natural perchlorate-reducing system as a pre-inoculant for mitigating perchlorate toxins in Martian regolith simulant. We will also evaluate the phased approach of introducing pioneer species and plant beneficial micro-organisms as an intermediate structure-building step for transforming Martian regolith into a viable agricultural substrate that will reduce plant stress and increase seed germination rates and overall biomass production. The results of this study will pave the way for long-term sustainable crop production in a Martian habitat. The Phase 1 laboratory setup and all major instruments (including the anaerobic chamber) have been calibrated. We've researched and tested rapid 16 sequencing of Phase 1 samples using the Oxford Nanopore Flongle sequencing platform, as well as best practices for analysis of Nanopore 16S and shotgun data using rapid kits and compared datasets from mock microbial communities. Sediment samples from the Pilot Valley (PV) field site have been collected for initiation of the Phase 1 experiment and are currently being pre-conditioned for the initiation of Phase 1. The Phase 2 and 3 growing system has been set up and calibrated to successfully establish ideal conditions for the growth of lettuce. In addition to lettuce, pilot growouts of identified pioneer species have been conducted to assess potential water flow changes induced by these pioneer species in the regolith and to see which one of these species would be the most reliable initial colonizer.
Bibliography Type:	Description: (Last Updated: 11/21/2023)
Abstracts for Journals and Proceedings	Lopez J.M, Lynch KL, Malo KE. "Plants, planets, and perchlorates." Texas Area Planetary Science Meeting, San Antonio, TX, August 17-18, 2023. Lightening Talk Abstracts.Texas Area Planetary Science Meeting, San Antonio, TX, August 17-18, 2023. https://sites.google.com/view/tapsmeeting/home , Aug-2023
Abstracts for Journals and Proceedings	Lynch KL, Simpson AC, Loureiro RR, Johnson T, Kovtun RN, Lopez JM, Malo KE, Rampe E, Rey K, Santiago-Vazquez L, Thorpe MT, Williams A, Jackson WA. "Perchlorate bioremediation for Martian regolith-based agriculture using natural microbial consortia from Mars analog sites on Earth." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023